Bare Demo of IEEEtran.cls for IEEE TRANSACTIONS ON MAGNETICS

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***Resumen*—This document presents a demonstration paper for submissions to the *SET International Journal of Broadcast Engineering* (SET IJBE). It provides authors with a practical example of the journal manuscript structure, including title, author information, abstract, keywords, sections, figures, tables, equations, lists, references, and acknowledgments. The sample text also explains the two submission paths available to contributors: direct submission to SET IJBE and submission through the annual SET Expo Congress R&D Sessions. The content is intentionally generic and does not report original technical results; instead, it illustrates formatting conventions, writing organization, and the principal elements expected in a complete manuscript. This abstract is limited to fewer than 150 words, in accordance with the journal guidelines and intended to guide authors in preparing compliant, consistent, and publication-ready submissions. It may also be used as a reference when adapting the template to actual research articles.**

***Index Terms*—broadcast engineering, manuscript template, article preparation, scientific publishing, IEEE style**

I. INTRODUCTION

THIS demo paper illustrates the structure and style expected for manuscripts submitted to the *SET International Journal of Broadcast Engineering* (SET IJBE), including Extended Abstracts and Final Papers.

SET IJBE is an open-access, peer-reviewed international scientific journal dedicated to communications engineering in the field of broadcasting. Its purpose is to disseminate relevant research findings and state-of-the-art technologies in broadcast and media engineering.[1] Accepted articles are published online on a one-article-at-a-time basis after final approval by the Editorial Board. In addition, printed issues are published twice a year and compile the articles published online during the preceding six months.

SET IJBE offers two submission paths. One option is submission through the Call for Papers of the annual SET Expo Congress R&D Sessions, in which authors initially submit an Extended Abstract of three to four pages. Alternatively, full papers may be submitted directly to SET IJBE, independently of the Congress. In both cases, this template should be used.

For Congress-related submissions, the official Call for Papers provides detailed information on topics of interest, required materials, deadlines, presentation requirements, and final paper categories. Additional information is available at https://set.org.br/events/setexpo?lang=en.

This template is intended only as a formatting example and does not replace standard references on technical writing and LaTeX usage, such as [1], [2].

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[1]More information is available at https://revistas.set.org.br/ijbe.

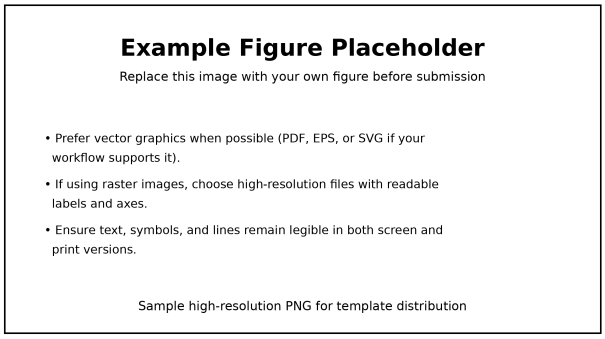

Fig. 1. Example of a figure caption.

II. MANUSCRIPT ELEMENTS

This section illustrates common elements of SET IJBE papers, including figures, tables, equations, lists, and references. The examples are provided only to demonstrate formatting and should be replaced with the actual content of the submission.

II-A. Figures

Figures should be clear, legible, and cited in the text, as in Fig. 1. Captions should be placed below the figure. Authors are encouraged to use vector graphics whenever possible. When raster images are used, high-resolution files should be provided so that labels, axes, and symbols remain readable in both electronic and printed versions. Examples of figures and graphical results are common in media and video research papers, such as those in video coding and quality assessment studies [3]–[5].

II-B. Tables

Tables should be numbered and cited in the text, as in Table I. In IEEE style, captions are usually placed above the table. Authors are encouraged to keep table formatting simple, clear, and consistent throughout the manuscript.

II-C. Equations

Mathematical expressions should be centered and numbered consecutively when they are referenced in the text. For example, (1) illustrates the format commonly used for displayed equations in technical papers. Similar mathematical notation is commonly found in signal processing, coding, and quality assessment literature [3]–[5].

$$E = mc^2 \tag{1}$$

II-D. Lists

Bulleted or enumerated lists may be used when appropriate to improve clarity. For example, the main parts of a technical contribution may be summarized as follows:

- presentation of the problem;
- description of the proposed approach; and
- discussion of the main results.

Código 1. Calcular el área y perímetro de un círculo.

```python
import numpy as np
while True:
    diametro_texto = input("Ingrese el valor del diámetro del círculo: ").strip()
    if not diametro_texto:
        print("Error: El valor no puede estar vacío.")
        continue
    try:
        diametro = float(diametro_texto)
        if diametro <= 0:
            print("Error: El diámetro debe ser un número positivo.")
            continue
        break
    except ValueError:
        print("Error: El valor ingresado no es un número válido.")
radio = diametro / 2
area = np.pi * radio ** 2
perimetro = 2 * np.pi * radio
decimales = 2
print(f"\nDiámetro: {diametro:.{decimales}f}")
print(f"Radio: {radio:.{decimales}f}")
print(f"Área del círculo: {area:.{decimales}f}")
print(f"Perímetro del círculo: {perimetro:.{decimales}f}")
```

II-E. References

References should be cited in the text using IEEE style, as illustrated by [1], [2]. All entries in the reference list should be complete, accurate, and consistently formatted according to the journal requirements. Examples of journal articles and technical references relevant to broadcast and media engineering are given in [3]–[6].

Tabla I
EXAMPLE OF A TABLE CAPTION

Item	Parameter	Factor	Value
A	Sample 1	1	123456
B	Sample 2	2	567890
C	Sample 3	7	142857

III. Conclusion

This demo paper has illustrated the basic structure and formatting expected for manuscripts submitted to the *SET International Journal of Broadcast Engineering* (SET IJBE). The sample text has shown how to prepare typical elements of a submission, including the title, author information, abstract, keywords, sections, figures, tables, equations, lists, references, and acknowledgment section.

Authors are encouraged to replace all sample text with their actual material and to verify that the final manuscript complies with the requirements of SET IJBE.

Apéndice A
Example of an Appendix

Appendices may be used to provide supplementary material that is relevant to the paper but would interrupt the flow of the main text if included in the body of the manuscript. Typical examples include detailed derivations, additional tables, implementation notes, or complementary experimental results.

As an illustration, suppose that an intermediate result used in the main text is given by

$$y[n] = \sum_{k=0}^{M-1} h[k]x[n-k]. \tag{2}$$

Equation (2) represents a discrete-time convolution and is included here only as a formatting example.

Authors should include appendices only when they add useful information for the reader. If no appendix is needed, this part of the template may simply be removed. If the paper is an Extended Abstract, appendices are not allowed.

Acknowledgment

The authors would like to thank the supporting institutions and collaborators who contributed to this work.

Referencias

[1] H. Kopka and P. W. Daly, *A Guide to LaTeX*, 3rd ed. Harlow, England: Addison-Wesley, 1999.

[2] L. Lamport, *LaTeX: A Document Preparation System*, 2nd ed. Boston, MA, USA: Addison-Wesley, 1994.

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